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(54) **NOSE PIECE FOR STRAND CUTTER**

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(57) **ABSTRACT**

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A strand cutter includes a drive assembly and a clamp assembly. The strand cutter includes a nose piece. The nose piece includes a nose piece body, a shoulder, a fixed blade receptacle and a rotating blade receptacle. The strand cutter includes a fixed blade received in the fixed blade receptacle and a rotating blade assembly. The rotating blade assembly includes a cylindrical cutting body, a replaceable shear blade or a cutting edge, and a drive gear engagement coupled to or integrated with the cutting body. The drive gear engagement engages the drive assembly so as to allow the transmission of torque from the drive assembly to the blade mount, the blade mount coupled to or integrally formed with the cutting body. The strand cutter also includes a flange plate, wherein the flange plate receives the shoulder. The flange plate retains the nose piece in engagement to the drive assembly.

Related U.S. Application Data

(60) Division of application No. 18/936,596, filed on Nov. 4, 2024, now Pat. No. 12,330,206, which is a continuation of application No. 18/807,358, filed on Aug. 16, 2024, which is a division of application No. 17/135,318, filed on Dec. 28, 2020, now Pat. No. 12,134,119.

(60) Provisional application No. 62/968,789, filed on Jan. 31, 2020, provisional application No. 63/006,202, filed on Apr. 7, 2020.

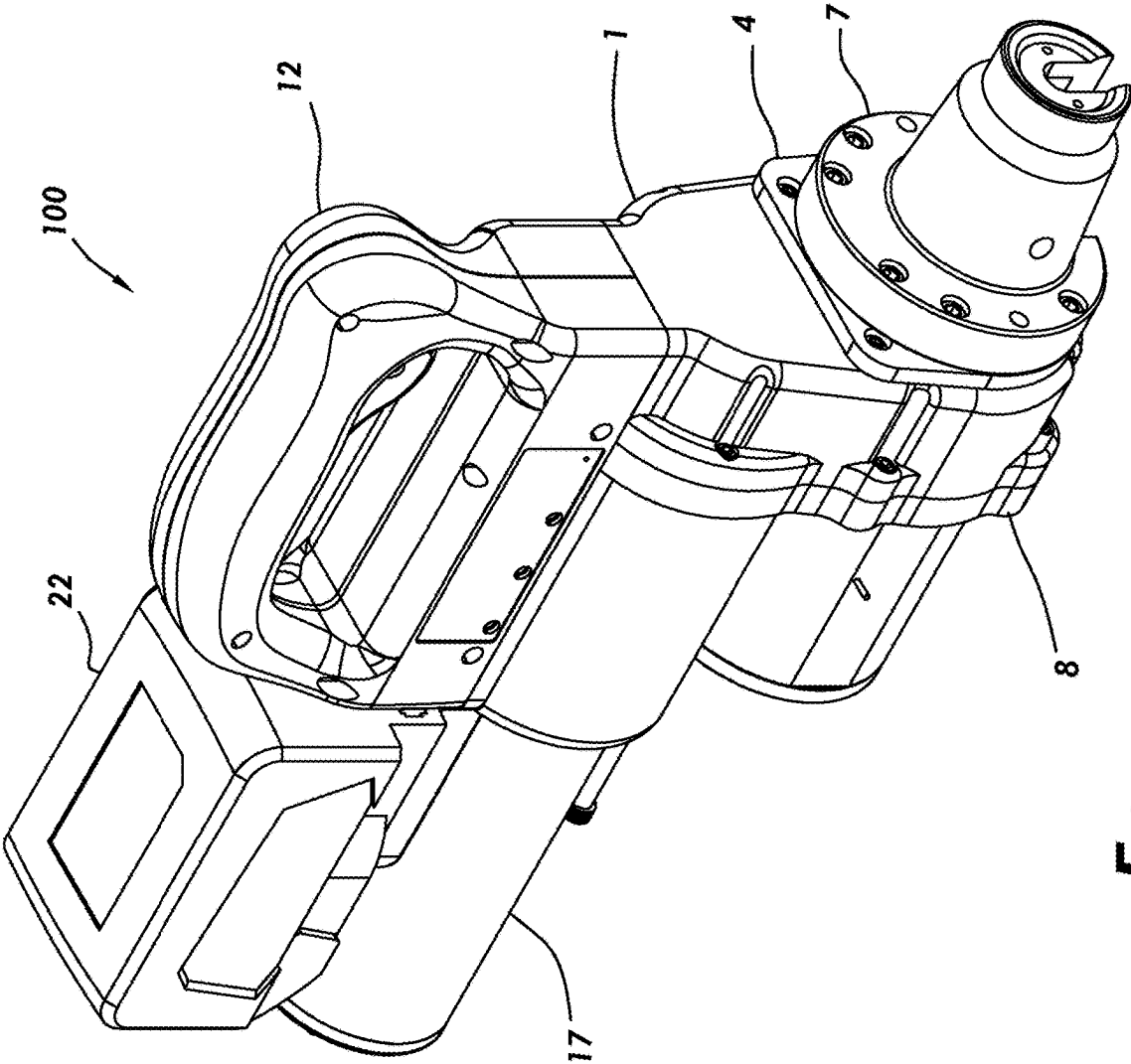
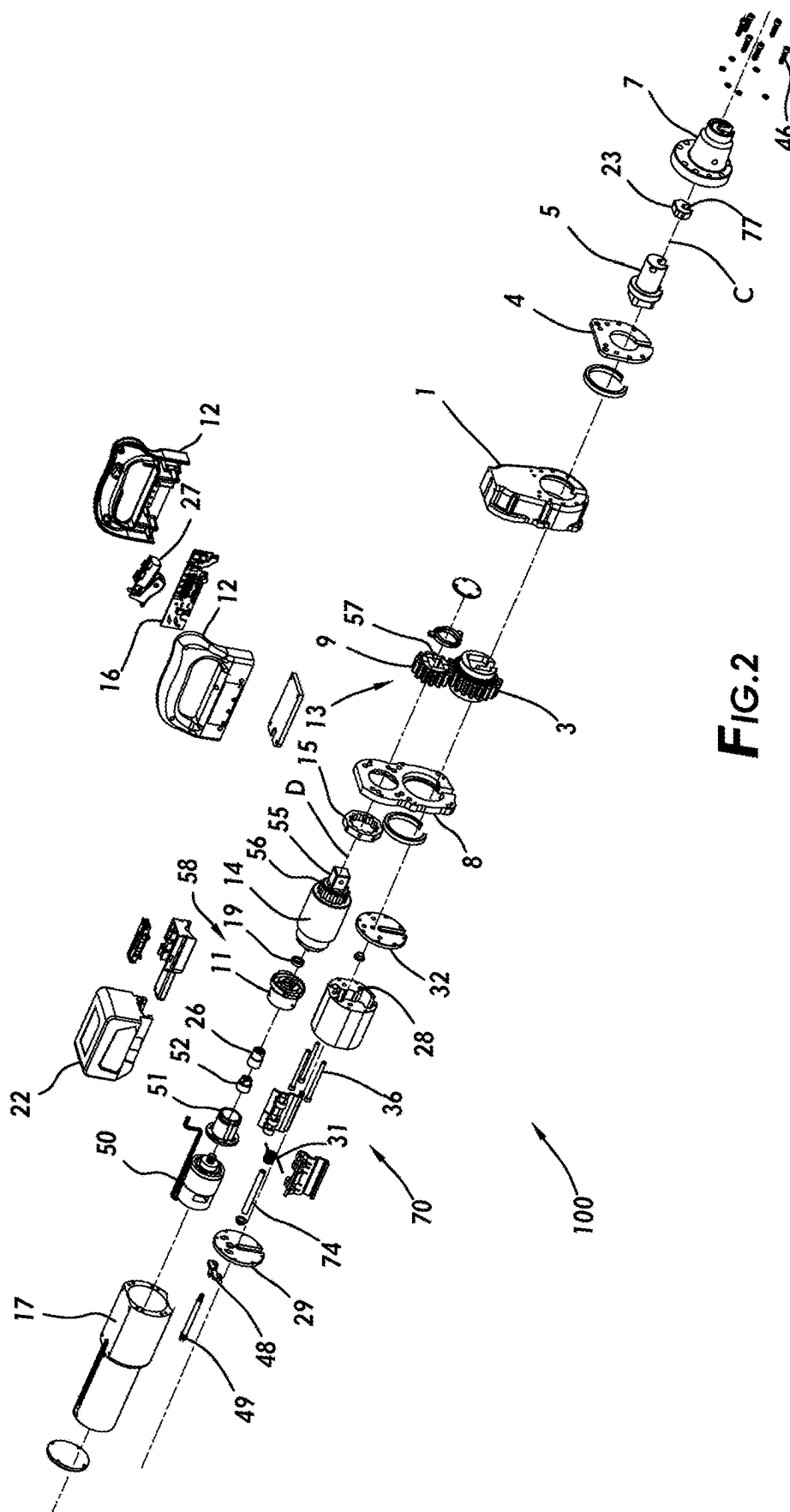


FIG. 1



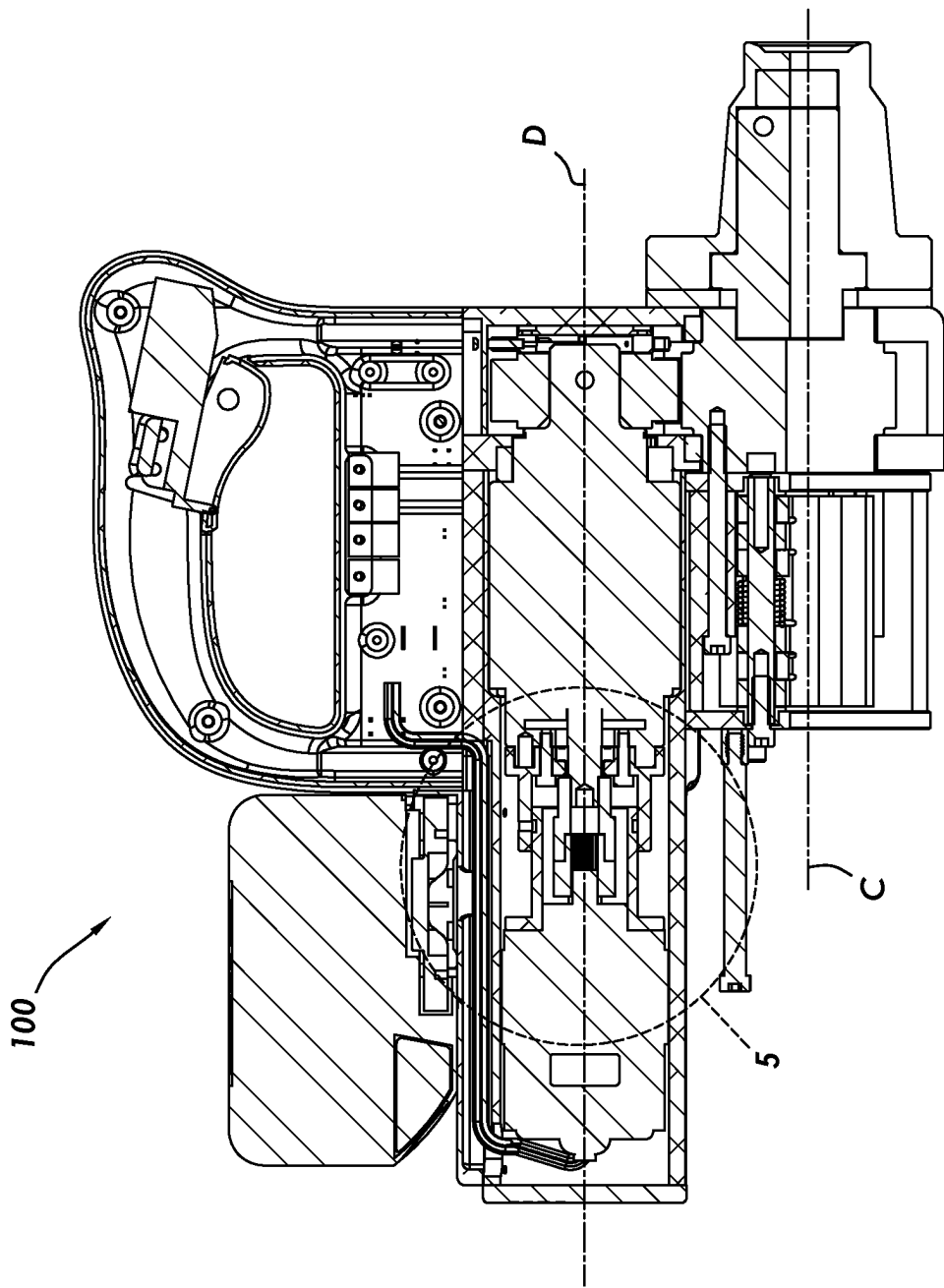


FIG.4

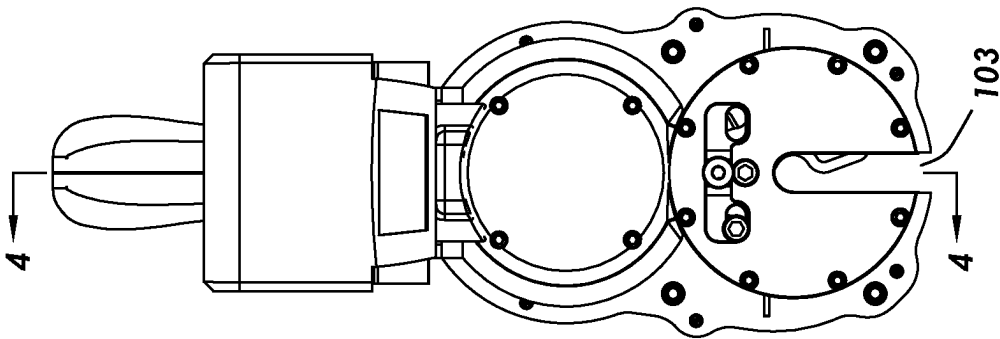


FIG.3

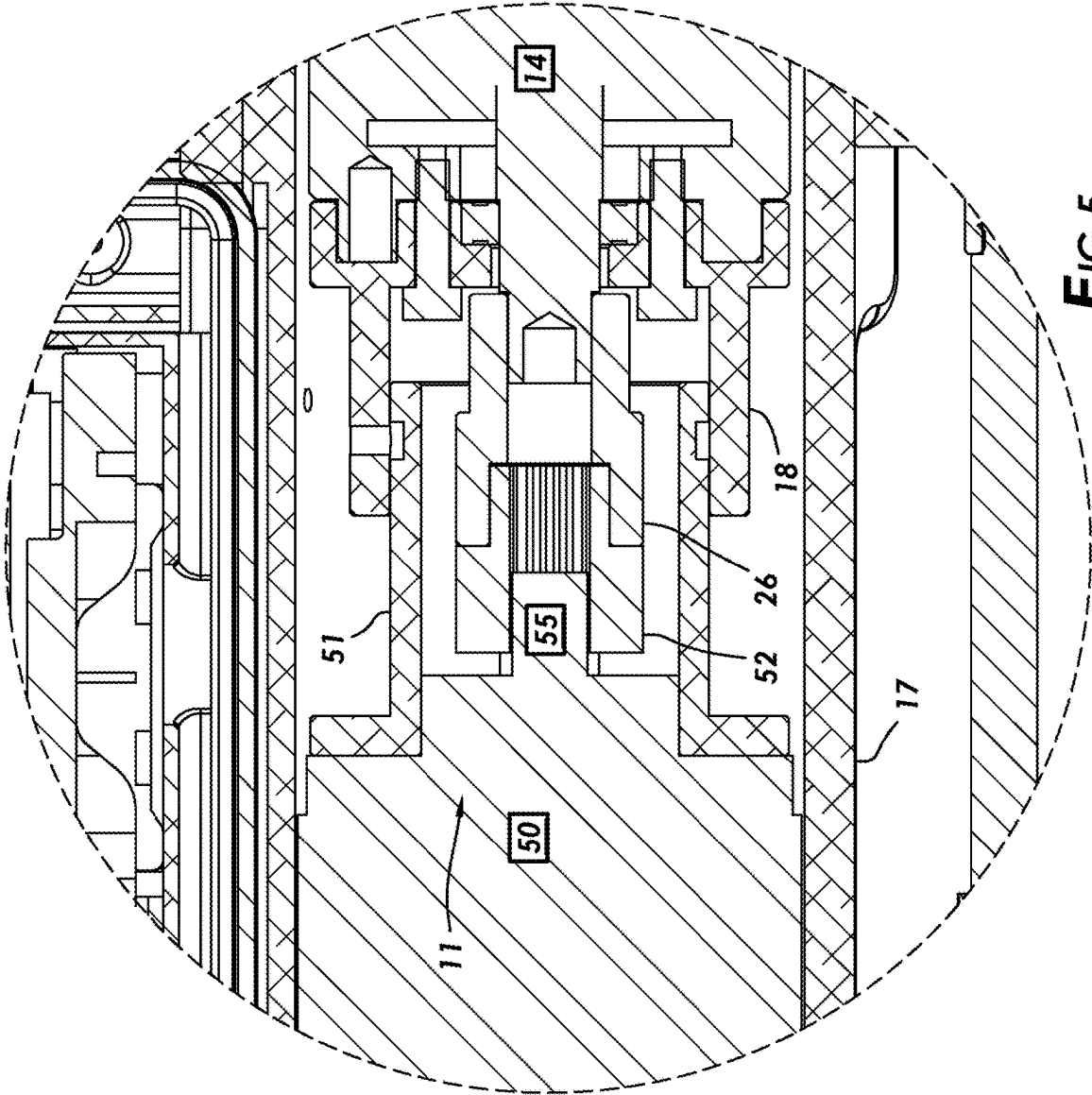


FIG. 5

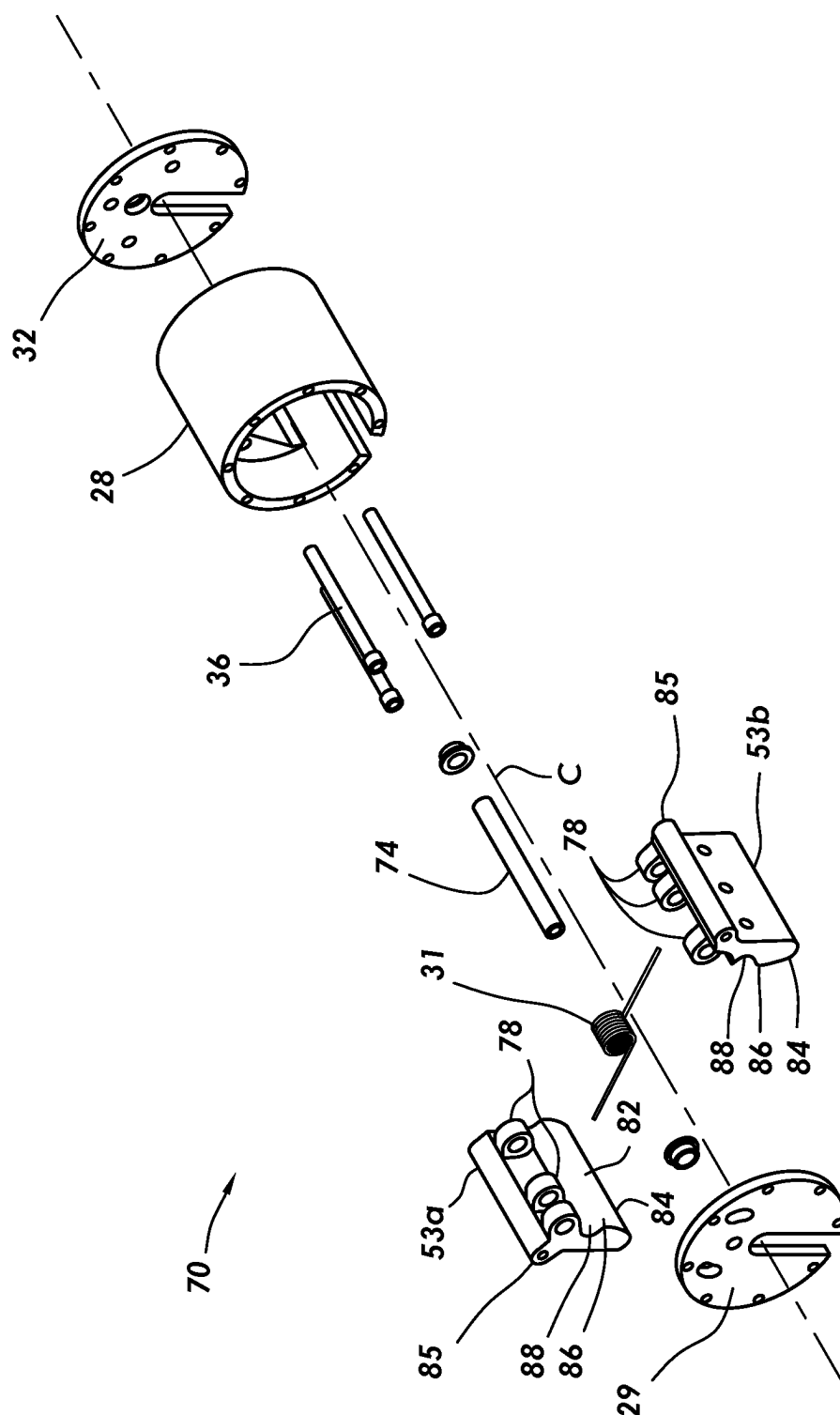


FIG. 6

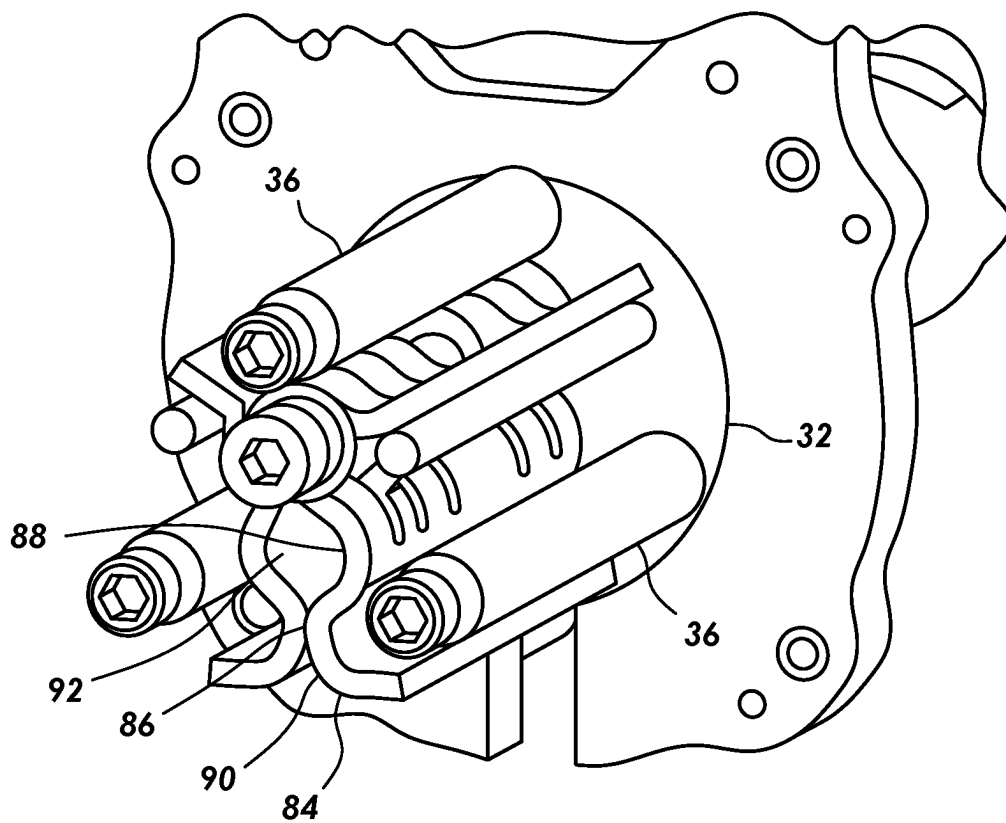


FIG. 7

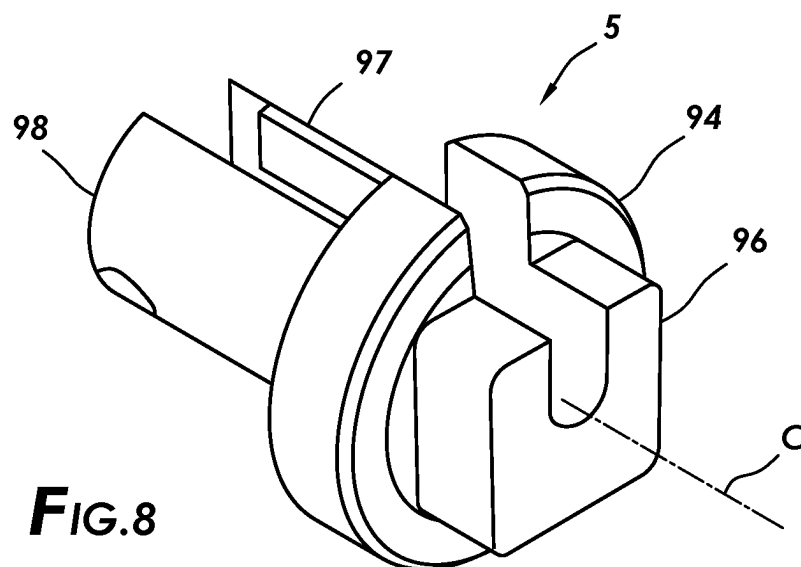
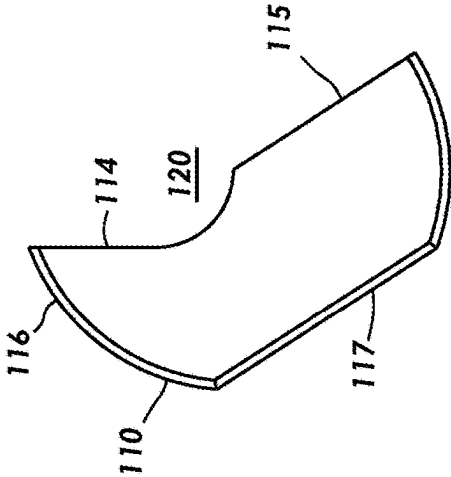
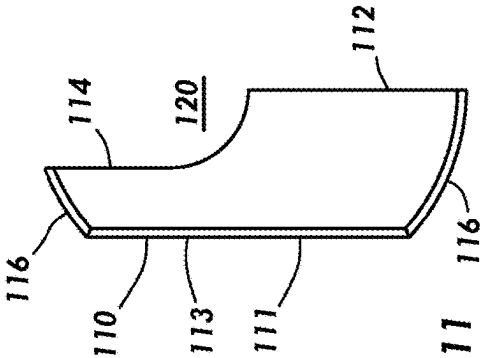
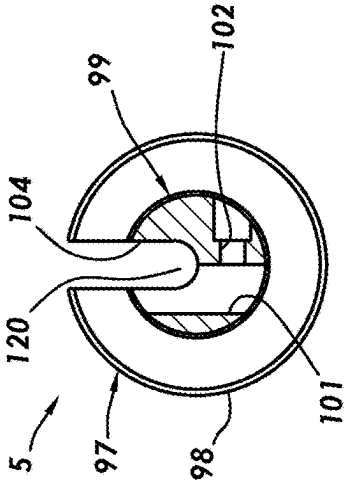
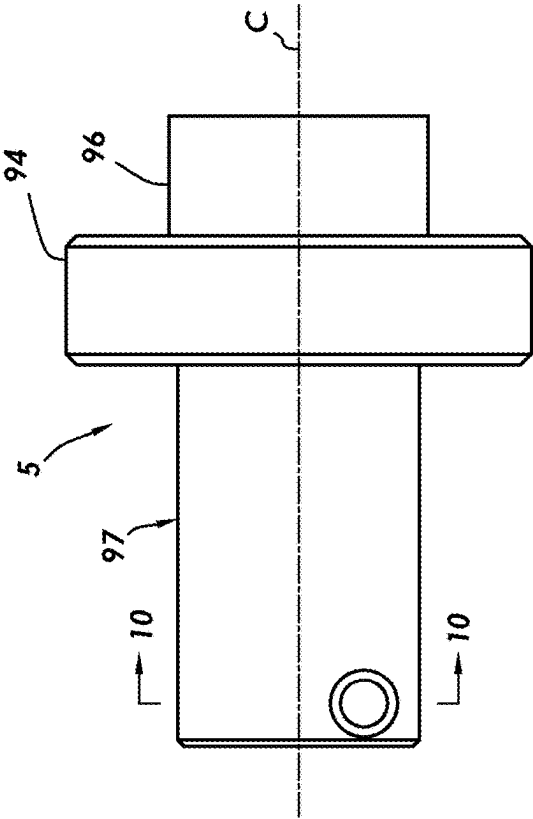


FIG. 8



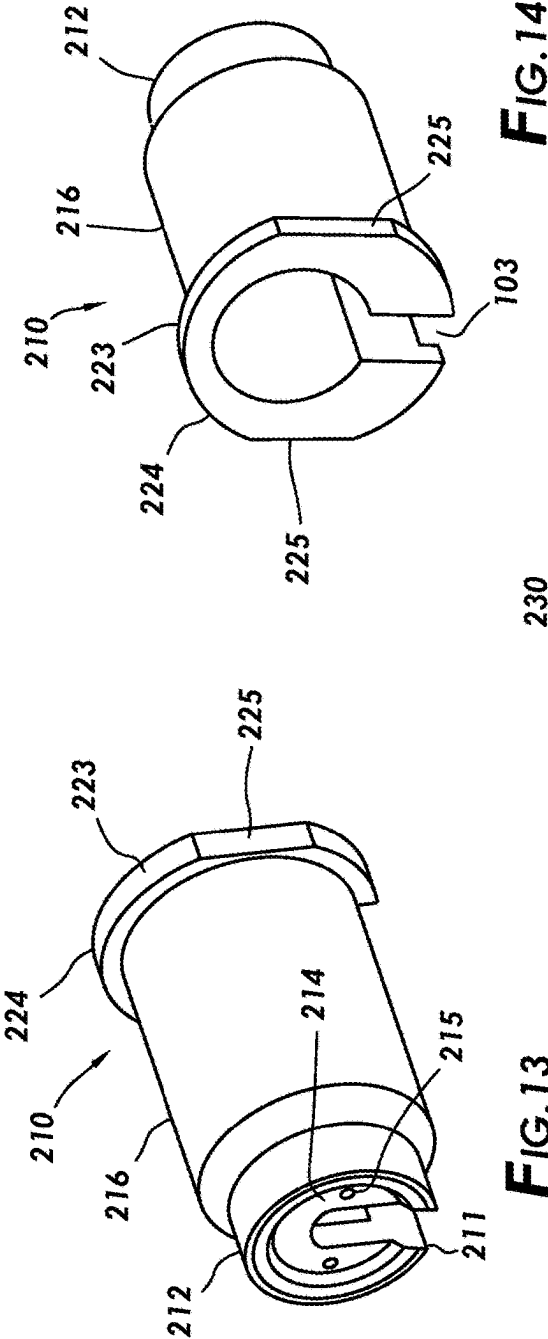


FIG. 14

FIG. 13

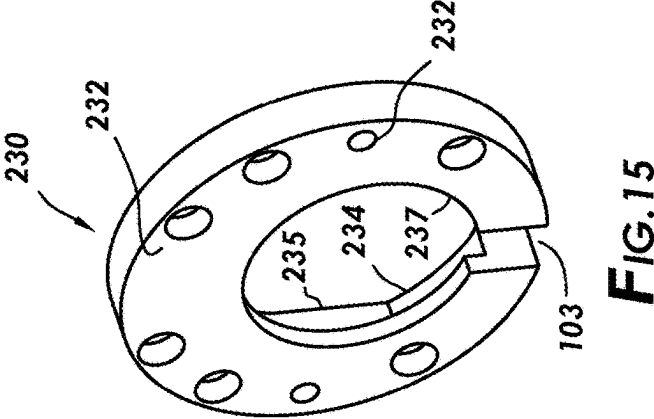


FIG. 15

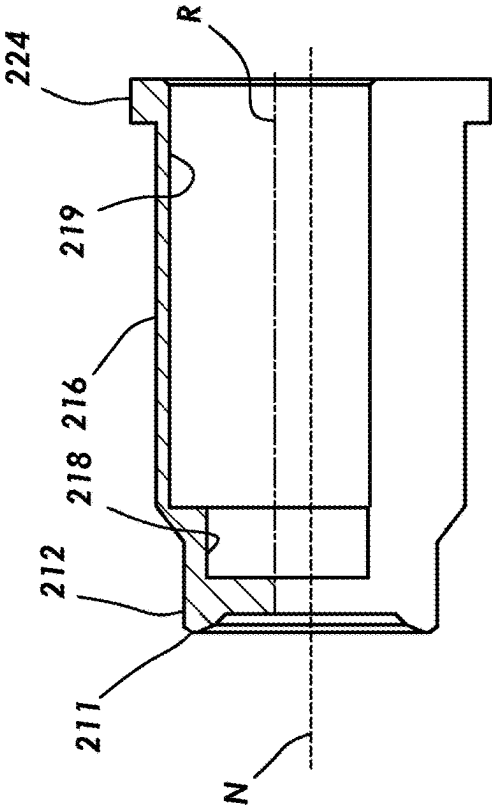


FIG. 16

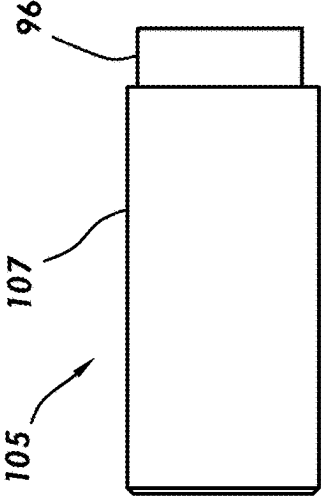


FIG. 17

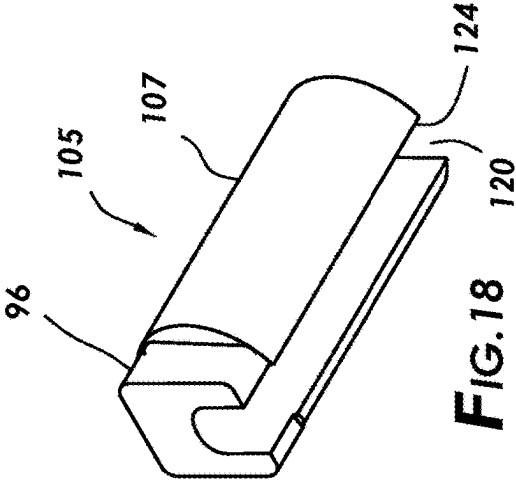


FIG. 18

NOSE PIECE FOR STRAND CUTTER

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a divisional application which claims priority to Ser. No. 18/936,596, filed Nov. 4, 2024, which is a continuation application of U.S. patent application Ser. No. 18/807,358, filed Aug. 16, 2024, which is a divisional application of U.S. patent application Ser. No. 17/135,318, filed on Dec. 28, 2020, now U.S. Pat. No. 12,134,119, that claims priority to U.S. provisional application No. 62/968,789, filed Jan. 31, 2020 and U.S. provisional application No. 63/006,202 filed Apr. 7, 2020, each of which is incorporated by reference herein in its entirety.

TECHNICAL FIELD/FIELD OF THE DISCLOSURE

[0002] The present disclosure is directed to a device for shearing a cable.

BACKGROUND OF THE DISCLOSURE

[0003] Many structures are built using concrete, including, for instance, buildings, parking structures, apartments, condominiums, hotels, mixed-use buildings, casinos, hospitals, medical buildings, government buildings, research/academic institutions, industrial buildings, malls, bridges, pavement, tanks, reservoirs, silos, foundations, sports courts, and other structures.

[0004] Pre-stressed concrete is structural concrete in which internal stresses are introduced to reduce potential tensile stresses in the concrete resulting from applied loads. This can be accomplished by two methods—pre-tensioned pre-stressing and post-tensioned pre-stressing. When post-tensioning concrete, the pre-stressing assembly is tensioned after the concrete has cured to a specified strength. The pre-stressing assembly, commonly known as a tendon, may include for example and without limitation, anchorages, one or more strands, and sheaths or ducts surrounding the strand(s). A strand may be tensioned between a pair of anchors, which are embedded in the concrete. The strand may be formed from a metal or composite or any suitable material exhibiting tensile strength, including, for example and without limitation, reinforcing steel, single wire cable, or multi-wire cable. The strand is typically fixedly coupled to a fixed anchorage positioned at one end of the tendon, the so-called “fixed end”, and is adapted to be stressed at the other anchor, the “stressing end” of the tendon. The strand is generally retained in each anchor by one or more wedges that engage a tapered recess in the anchor body so that when the strand is placed under tension, the wedges engage the strand more tightly.

[0005] One end of the strand extends through the stressing end anchor and out of the concrete body in which the anchor is embedded. In some cases, a pocket former is placed around the strand before the concrete is poured, which results in a pocket in the cured concrete. In these cases, the end of the strand extends through and outwardly from the pocket. Once the concrete has cured, a tensile force can be applied to the extending strand end, causing an elongation of the strand. Releasing the tensile force causes the strand to be more tightly gripped by the wedges, thereby maintaining its elongated stress, which is transferred to the concrete via the

anchors. The portion of the strand that extends out of the stressing anchor is typically removed.

SUMMARY

[0006] The present disclosure provides for a strand cutter. The strand cutter includes a drive assembly and a clamp assembly mechanically coupled to the drive assembly. In addition, the strand cutter includes a nose piece, wherein the nose piece includes a nose piece body, a shoulder, a fixed blade receptacle and a rotating blade receptacle. Further, the strand cutter includes a fixed blade received in the fixed blade receptacle and a rotating blade assembly. The rotating blade assembly includes a cylindrical cutting body, a replaceable shear blade or a cutting edge, and a drive gear engagement coupled to or integrated with the cutting body. The rotating blade assembly is received in the rotating blade receptacle. The drive gear engagement engages the drive assembly so as to allow the transmission of torque from the drive assembly to the blade mount, the blade mount coupled to or integrally formed with the cutting body. The strand cutter also includes a flange plate, wherein the flange plate receives the shoulder, wherein the flange plate retains the nose piece in engagement to the drive assembly.

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] The present disclosure is best understood from the following detailed description when read with the accompanying figures. It is emphasized that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0008] FIG. 1 is an isometric view of a strand cutter in accordance with an embodiment of the disclosure.

[0009] FIG. 2 is an isometric exploded view of the strand cutter of FIG. 1.

[0010] FIG. 3 is an end view of the strand cutter of FIG. 1.

[0011] FIG. 4 is a cross-section along lines 4-4 of FIG. 3.

[0012] FIG. 5 is an enlarged view of a portion of FIG. 4.

[0013] FIG. 6 is an isometric exploded view of a clamp assembly in accordance with an embodiment of the disclosure.

[0014] FIG. 7 is an isometric view of a portion of an assembled clamp assembly in accordance with an embodiment of the disclosure.

[0015] FIG. 8 is an isometric view of a rotating blade assembly in accordance with an embodiment of the disclosure.

[0016] FIG. 9 is a side view of the rotating blade assembly of FIG. 8.

[0017] FIG. 10 is a cross-section along lines 10-10 of FIG. 9.

[0018] FIG. 11 is an enlarged view of a replaceable shear blade in accordance with an embodiment of the disclosure.

[0019] FIG. 12 is an enlarged view of a replaceable shear blade in accordance with an embodiment of the disclosure.

[0020] FIGS. 13 and 14 are isometric front and rear views, respectively, of a nose piece in accordance with an embodiment of the disclosure.

[0021] FIG. 15 is an isometric view of a flange plate for use with the embodiment of FIGS. 13 and 14.

[0022] FIG. 16 is a cross-sectional side view of the nose piece of FIG. 13.

[0023] FIGS. 17 and 18 are side and isometric views of a rotating blade assembly for use with an alternate embodiment of the disclosure.

DETAILED DESCRIPTION

[0024] It is to be understood that the following disclosure provides many different embodiments, or examples, for implementing different features of various embodiments. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0025] The present disclosure provides a strand cutter for shearing post-tensioning cables, for example, in the field. Thus, the strand cutter is designed to operate adjacent to a post-tensioning anchor or within a pocket formed adjacent to a post-tensioning anchor. The strand is cut by receiving the strand eccentrically in a rotating blade assembly and rotating a replaceable shear blade relative to a fixed blade

[0026] Referring to FIGS. 1 and 2, strand cutter 100 in accordance with the disclosure may include main drive housing 17. The top side of main drive housing 17 may support battery 22 and tool handle 12. Tool handle 12 may include trigger 27 and printed circuit board 16. Printed circuit board 16 may control the operation of strand cutter 100 in response to inputs that may include signals from trigger 27. Drive assembly 58, i.e., electric motor 50 and gear box 14 may be positioned within main drive housing 17. Electric motor 50 and gearbox 14 may be mechanically coupled and coaxial within main drive housing 17.

[0027] Strand cutter 100 may further include front housing 1 that is mechanically coupled to main drive housing 17. Front housing 1 may be affixed to gearbox 14 via mounting plate 8. Wear plate 4 may be mounted to the front of front housing 1 on a first side of wear plate 4. The second side of wear plate 4 may be mounted to nose piece 7. Nose piece 7, further described below, may define a cutting chamber. Front housing 1 may house primary drive gear 9 and secondary drive gear 3.

[0028] Referring to FIGS. 4 and 5, drive assembly 58 for strand cutter 100 may be housed in main drive housing 17 and may include electric motor 50, gearbox 14, and gearbox connector assembly 11 that couples electric motor 50 to gearbox 14. Gearbox connector assembly 11 may include motor mount adaptor 51, motor shaft coupler 52, gearbox shaft coupler 26, gearbox to motor mount 18, and bearing 19.

[0029] Referring again to FIG. 2, male gearbox spline 56 and coaxial output shaft 55 may extend from gearbox 14. Male gearbox spline 56 may engage female gearbox spline 15. Female gearbox spline 15 may have a hexagonal or other non-round perimeter so as to engage a corresponding receptacle in mounting plate 8. Engagement of male and female gearbox splines 56, 15, and engagement of female gearbox spline 15 with mounting plate 8 may prevent drive assembly 58 from rotating relative to other components of strand cutter 100.

[0030] Drive gears 13, including primary drive gear 9 and secondary drive gear 3, transmit torque from electric motor 50 through gear box 14 to rotating blade assembly 5 (or 105). In certain embodiments, primary drive gear 9 may include drive opening 57 that corresponds to the shape of coaxial output shaft 55, allowing the transmission of torque from coaxial output shaft 55 to primary drive gear 9. Teeth on the perimeter of primary drive gear 9 engage corresponding teeth on the outside of secondary drive gear 3, allowing the transmission of torque from primary drive gear 9 to secondary drive gear 3. Rotating blade assembly 5 (or 105) is mechanically coupled to secondary drive gear 3 so as to be rotated thereby. Rotating blade assembly 5 may extend through openings in front housing 1, wear plate 4, and into the cutting chamber in nose piece 7. Fixed blade 23 is also received and non-rotatably retained in the cutting chamber of nose piece 7 adjacent to rotating blade assembly 5. Fixed blade 23 includes fixed strand slot 77.

[0031] Electric motor 50, motor shaft coupler 52, gearbox shaft coupler 26, gearbox to motor mount 18, gearbox 14, gearbox splines 56, 15, and primary drive gear 9 define drive axis D. Similarly, clamp housing 28, secondary drive gear 3, rotating blade assembly 5 and fixed blade 23 define cutting axis C.

Cutting Assembly

[0032] Referring now to FIGS. 8-10, strand cutter 100 may include rotating blade assembly 5. Rotating blade assembly 5 may include mounting shoulder 94, drive gear engagement 96 mounted to mounting shoulder 94, and blade mount 97 mounted to mounting shoulder 94. Mounting shoulder 94 may be configured to rotate within front housing 1 and to support rotating blade assembly 5 within the strand cutter. Drive gear engagement 96 may be non-circular and may be configured to engage a corresponding opening in secondary drive gear 3, so as to allow the transmission of torque from secondary drive gear 3 to rotating blade assembly 5. Blade mount 97 may be generally cylindrical and may comprise an outer wall 98 and an inner shaft 99. Inner shaft 99 may include blade receptacle 101, pin opening 102, and strand slot portion 104. Blade receptacle 101 may receive replaceable shear blade 110. Pin opening 102 may receive a set screw or other fastener to retain replaceable shear blade 110 within blade receptacle 101. Strand slot portion 104 may align with throat 103.

[0033] Referring now to FIG. 11, replaceable shear blade 110 in accordance with some embodiments may include first drive face 112, second drive face 113, curved outer surface 116, and blade edge 114. Second drive face 113 and curved outer surface 116 may include chamfered edge 111. Drive faces 112, 113 engage corresponding surfaces in blade receptacle 101. Drive faces 112, 113 transmit torque from inner shaft 99 to replaceable shear blade 110. Inner shaft 99 may transmit a cutting torque to first drive face 112 (clockwise as shown in FIG. 11) during a cutting operation and may transmit an opening torque (counter-clockwise in FIG. 11) to second drive face 113 after a cut has been completed. In the illustrated embodiment, drive faces 112, 113 are planar and parallel, but drive faces 112, 113 may have any suitable configuration.

[0034] By way of example and without limitation, in some embodiments, replaceable shear blade 110 may have drive faces 115, 117, that are not parallel to cutting strand slot 120 and instead define an angle between 5 and 65 degrees with

cutting strand slot **120**, as shown in FIG. **12**. In still other embodiments, replaceable shear blade **110** may have one drive face **115** or **117** instead of two. In some embodiments drive face **115** or **117** may extend between the bottom of cutting strand slot **120** and the outer circumference of inner shaft **99**, as shown in FIGS. **11** and **12**. In other embodiments drive face **115** or **117** may not extend to the outer circumference of inner shaft **99** or may not be a straight line.

[0035] Curved outer surface **116** may match the outer surface of outer wall **98** so as to provide a substantially continuous surface. To assemble rotating blade assembly **5**, replaceable shear blade **110** may be positioned in blade receptacle **101** and secured with a fastener in pin opening **102**. Blade edge **114** cooperates with strand slot portion **104** of inner shaft **99** to define a cutting strand slot **120** (FIG. **10**) that is aligned with strand channel **92** and fixed strand slot **77** and may receive a strand for cutting.

[0036] Because replaceable shear blade **110** is removable from rotating blade assembly **5**, replaceable shear blade **110** may be replaced when worn or damaged. Likewise, replaceable shear blade **110** may be selected from a variety of replaceable shear blades **110** of different materials or hardnesses, depending on the cutting application. In some embodiments, replaceable shear blade **110** may be accessed by removing nose piece **7** and removing the pin in pin opening **102**.

[0037] U-shaped strand slots **120**, **77** of rotating blade assembly **5** and fixed blade **23**, respectively, are configured to receive a strand (cable) therein. The “bottom” or terminus of each U-shaped strand slot may be adjacent to cutting axis C. Fixed blade **23** and rotating blade assembly **5** are configured such that a strand positioned in the respective strand slots will be sheared when rotating blade assembly **5** has rotated 180° about cutting axis C. When cutting strand slot **120** of rotating blade assembly **5** is aligned with fixed strand slot **77** of fixed blade **23**, i.e. at 0° of rotation, the tool is in a fully open position.

Clamp Assembly

[0038] Referring to FIGS. **2** and **6**, the strand cutter **100** may also include clamp assembly **70**. Clamp assembly **70** is mechanically coupled to the drive assembly **58**. In certain embodiments, clamp assembly **70** may include clamp housing **28** that is enclosed at one end by clamp cover plate **29** and at the other end by clamp mount plate **32**. Housed within clamp housing **28** may be left and right clamp arms **53a**, **53b**, respectively (FIG. **6**), pivot rod **74**, torsion spring **31**, and at least one clamp mounting screw **36**. Clamp mounting screw(s) **36** may extend longitudinally through clamp housing **28** and through clamp mount plate **32**. In some embodiments, the forward end(s) of clamp mounting screw(s) **36** may engage secondary drive gear **3** to allow clamp assembly **70** to rotate about cutting axis C in conjunction with rotating blade assembly **5**.

[0039] Left and right clamp arms **53a**, **53b** each include shoulder **85** and one or more sleeves **78**. In the illustrated embodiment, each clamp arm **53a**, **53b** has three sleeves **78**, but the number of sleeves is a matter of design choice. Sleeves **78** are configured so that left and right clamp arms **53a**, **53b** may be positioned in an operational configuration in which sleeves **78** are coaxial and define a pivot axis. Pivot rod **74** may extend through sleeves **78**. Torsion spring **31** may be positioned around pivot rod **74** such that one end of torsion spring **31** bears on each clamp arm **53a**, **53b**, thereby

urging clamp arms **53a**, **53b** in opposite directions relative to pivot rod **74**, resulting in a spring-loaded clamp assembly **70**. Sleeves **78** may also be longitudinally spaced such that sleeves **78** on left clamp arm **53a** interleave with sleeves **78** on right clamp arm **53b**, allowing left and right clamp arms **53a**, **53b** to be longitudinally aligned.

[0040] In addition to sleeves **78**, left and right clamp arms **53a**, **53b** each include curved clamping face **82** that includes guide **84**, neck portion **86**, and strand-receiving groove **88**. Each neck portion **86** may be positioned between guide **84** and strand-receiving groove **88**. Referring to FIG. **7**, when clamp assembly **70** is in an operational configuration, clamping faces **82** are urged toward each other by torsion spring **31**. Guides **84** cooperate to define clamp entry **90** and strand-receiving grooves **88** cooperate to define strand channel **92**. In some embodiments, neck portions **86** may bear on each other. Clamp entry **90** and strand channel **92** may be parallel with cutting axis C.

[0041] Strand cutter **100** may cut a strand at a predetermined point on the strand without requiring the strand to be threaded into a cutting device. Strand cutter **100** is provided with a throat **103**, such as shown in FIG. **3**. Throat **103** may include a groove or slot that extends along a bottom surface of the strand cutter **100**. Throat **103** provides for a strand to be received into clamp assembly **70** and strand slots **77**, **120**, laterally, i.e. by moving strand cutter **100** in a direction normal to the strand and normal to cutting axis C, so that cutting axis C moves toward the strand, rather than threading the strand through a closed opening along cutting axis C.

[0042] In some embodiments, throat **103** may be formed by providing a strand-receiving slot in each of the components along cutting axis C. Thus, clamp assembly **70**, mounting plate **8**, secondary drive gear **3**, front housing **1**, wear plate **4**, rotating blade assembly **5**, fixed blade **23**, and nose piece **7** may each include a strand-receiving slot. For clamp assembly **70**, clamp housing **28**, clamp cover plate **29**, and clamp mount plate **32** each include a strand-receiving slot, so that a strand may enter clamp assembly **70** laterally. Similarly, in mounting plate **8**, secondary drive gear **3**, front housing **1**, wear plate **4**, and nose piece **7**, the respective strand-receiving slots are configured to receive a strand and to align with the strand slots when the tool is in an open position. For fixed blade **23** and rotating blade assembly **5**, the strand-receiving slots are shown as strand slots **77** and **120**, i.e. the slots whose relative rotation results in cutting of the strand. The throat-forming strand-receiving slots are aligned and configured such that a strand may be positioned on cutting axis C by placing strand cutter **100** onto a strand laterally.

[0043] In some embodiments, as strand cutter **100** is placed on a strand, the strand enters throat **103**. As the strand moves into spring-loaded clamp assembly **70**, the strand will enter clamp entry **90** and will bear on neck portions **86**. Applying to strand cutter **100** an engagement force normal to the strand will cause the strand to push neck portions **86** apart. Application of a sufficient engagement force will overcome the opposing force of torsion spring **31**, allowing clamping arms **53a**, **53b**, to rotate about pivot rod **74** and allow the strand to enter strand channel **92**. Once the strand is received in strand channel **92**, torsion spring **31** urges clamping arms **53a**, **53b**, to rotate back to a closed position in which the strand is retained in strand channel **92** by neck portions **86**.

[0044] In some embodiments, because no separate operation is required in order to fix a strand in strand cutter 100 in a preparation for cutting, an operator's hands can be used for positioning and operating the strand cutter and can remain clear of the strand and the cutting zone.

[0045] In some embodiments, a clamp opener 48 may work in conjunction with pin 49 that extends through clamp housing 28 into engagement with clamp arms 53a and/or 53b to enable clamp assembly 70 to be opened manually instead of by applying an opening force via a strand.

[0046] Referring now to FIGS. 13-16, in certain embodiments, nose piece 7, as shown in FIGS. 1 and 2, is replaced with a two-piece nose piece assembly that includes generally cylindrical nose piece 210 and flange plate 230, i.e., the combination of FIGS. 13 and 15. Nose piece 210 may include pocket engagement section 212 coupled or formed integrally with nose piece body 216, which is coupled to or formed integrally with shoulder 224. Pocket engagement section 212 may include forward lip 211 and recessed face 214. Pocket engagement section 212 may also include fixed blade receptacle 218 for receiving fixed blade 23 (FIG. 2). Pocket engagement section 212 may also include one or more extraction holes 215 extending through recessed face 214 and into fixed blade receptacle 218 for facilitating removal of fixed blade 23 from fixed blade receptacle 218. The inner wall of fixed blade receptacle 218 and the outside of fixed blade 23 correspond in shape and may each have a square, hexagonal, or other non-round shape to facilitate the transmission of torque to fixed blade 23 and may prevent rotation of fixed blade 23 within nose piece 210. Nose piece 210 has a longitudinal nose piece axis N.

[0047] Nose piece body 216 may be frustoconical or may be generally cylindrical as illustrated and may include an eccentric bore that defines a rotating blade receptacle 219. Rotating blade receptacle 219 may be generally cylindrical with a longitudinal axis R that is parallel to but offset from nose piece axis N. Rotating blade receptacle 219 is configured to receive and support a rotating blade assembly such as rotating blade assembly 5 or rotating blade assembly 105 described below.

[0048] Shoulder 224 of nose piece 210 may have generally circular circumferential face 223. Circumferential face 223 may have a diameter greater than the diameter of nose piece body 216 and may include at least one flat 225. In the embodiment illustrated in FIGS. 13 and 14, circumferential face 223 includes two parallel flats 225; in other embodiments shoulder 224 may include more than two flats 225 or may have a non-circular shape, such as square or hexagonal. Shoulder 224 may be integrally formed with nose piece 210.

[0049] Referring to FIG. 15, flange plate 230 may be generally circular and may include retaining face 232 having lip 234 extending longitudinally therefrom. The longitudinal extent of lip 234 may be substantially equal to the thickness of shoulder 224. Retaining face 232 may include eccentric opening 237 that is sized and configured to receive nose piece body 216 therethrough. Eccentric opening 237 may be sized to prevent or restrict the passage of shoulder 224 therethrough, whereby flange plate 230 may be used to retain nose piece 210 in a desired position. The inner surface of lip 234 may include flats 235 configured to correspond to and engage with flats 225 on nose piece 210. Engagement of flats 235 with 225 on nose piece 210 may help prevent rotation of nose piece 210 relative to flange plate 230. The combination of fasteners 46 affixing flange plate 230 to

strand cutter 100 and engagement of flats 235 with flats 225 serves to non-rotatably couple nose piece 210 to front housing 1 and to place rotating blade assembly 105 into engagement with drive assembly 58.

[0050] Flange plate 230 and nose piece 210 may each include a slot that is configured to align with slots in the other components of strand cutter 100 and form part of throat 103.

[0051] Referring now to FIGS. 17 and 18, in certain embodiments, rotating blade assembly 105 may include cutting body 107 coupled to or integrally formed with drive gear engagement 96 and blade mount 97. Rotating blade assembly 105 may be configured to rotate within and relative to rotating blade receptacle 219 in nose piece 210. Drive gear engagement 96 may be non-circular protuberance coupled to or formed integrally with cutting body 107 and engages a corresponding opening in secondary drive gear 3, so as to allow the transmission of torque from secondary drive gear 3 to rotating blade assembly 5. Cutting body 107 may be generally cylindrical. Cutting body 107 may in some embodiments include blade edge 124 or, in other embodiments, replaceable shear blade 110 as described above. Like rotating blade assembly 5, rotating blade assembly 105 may include cutting strand slot 120. Cutting strand slot 120 aligns with throat 103 when the tool is in an open position and ready to receive a strand therein.

[0052] To assemble, rotating blade assembly 105 may be placed into rotating blade receptacle 219 in nose piece 210. Nose piece 210 may be assembled into flange plate 230 such that nose piece body 216 extends through eccentric opening 237 and shoulder 224 is received in lip 234 such that flats 225 engage flats 235. The assembled components 230, 210, and 105 can be assembled with the other components of strand cutter 100 by affixing flange plate 230 to front housing 1.

Operation

[0053] To cut a strand with strand cutter 100, strand cutter 100 is placed onto the strand at the desired location along the strand. Throat 103 is aligned with the strand and strand cutter 100 is advanced toward the strand until the strand passes through the neck of clamp assembly 70 and is received in strand channel 92. Provided the components of strand cutter 100 are properly aligned and the tool is in the open configuration, the strand will also be received in strand slots 120 and 77.

[0054] Actuation of trigger 27 will cause power to flow from battery 22 to electric motor 50. Electric motor 50 will output torque that is transmitted via gearbox 14 and coaxial output shaft 55 to primary drive gear 9, which transmits the torque to secondary drive gear 3. Secondary drive gear 3 in turn transmits torque to rotating blade assembly 5 (or 105). In response, rotating blade assembly 5, including replaceable shear blade 110, rotates relative to strand cutter 100. Because fixed blade 23 is non-rotatably mounted to the device, rotating blade assembly 5, including replaceable shear blade 110, rotates relative to fixed blade 23, thereby shearing the strand. When rotating blade assembly 5 has rotated 180° relative to fixed blade 23, the strand will be completely cut. The strand cutter 100 can then be removed from the strand and the rotating blade assembly can be returned to the open position, in which cutting strand slot 120 aligns with throat 103.

[0055] In the event that any component in nose piece **210**, including fixed blade **23**, rotating blade assembly **5**, rotating blade assembly **105**, or nose piece **210** itself, breaks, becomes dull or otherwise needs replacement, flange plate **230** can be decoupled from strand cutter **100** and the necessary component replaced with a new component. Replacement is quick and easy and does not require disassembly of the rest of strand cutter **100**.

1-15. (canceled)

16. A blade assembly for a strand cutter, comprising:
a rotating shear blade assembly that comprises a drive gear engagement;
a rotatable blade mount for the rotating shear blade assembly;
a shear blade;

wherein the drive gear engagement is configured to engage a drive assembly so as to allow transmission of torque from the drive assembly to the rotatable blade mount, and further wherein the rotatable blade mount comprises a blade receptacle that is configured to receive the shear blade.

17. The blade assembly of claim **16**, wherein the shear blade includes a first drive face and a second drive face.

18. The blade assembly of claim **17**, wherein the first drive face and second drive face are planar and parallel.

19. The blade assembly of claim **16**, wherein the rotatable blade mount further includes a strand slot portion and wherein the shear blade cooperates with the strand slot portion to define a cutting strand slot.

20. The blade assembly of claim **16**, further comprising:
a fixed blade, wherein the fixed blade includes a fixed strand slot.

21. The blade assembly of claim **17**, wherein the rotatable blade mount comprises:

an inner shaft having an outer circumference,

a fixed blade including a fixed strand slot,

and the first drive face and the second drive face extends between a bottom of the fixed strand slot and the outer circumference of the inner shaft.

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